

From
an explicit to an **implicit** MOO model
with JuMP and MOA

Example (1/3)



$$\begin{aligned}
 \max z_1 &= x_1 + x_2 \\
 \min z_2 &= x_1 + 3x_2 \\
 \text{s.t.} \quad &2x_1 + 3x_2 \leq 30 \\
 &3x_1 + 2x_2 \leq 30 \\
 &x_1 - x_2 \leq 5.5 \\
 &x_1, x_2 \in \mathbb{N}
 \end{aligned}$$

$$\begin{aligned}
 \max z_k &= \sum_{j=1}^n c_j^k x_j \quad k = 1, \dots, p \\
 \text{s.t.} \quad &\sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\
 &x_j \in \mathbb{N} \quad j = 1, \dots, n
 \end{aligned}$$

```

julia> c1 = [1,1]; c2 = -1 * [1,3]; c = vcat(c1',c2')
julia> a = [2 3; 3 2; 1 -1]
julia> b = [30, 30, 5.5]
julia> m,n = size(a)
julia> p = size(c,1)
    
```

```

julia> md = Model( )
julia> @variable(md, x[1:n] ≥ 0, Int)
julia> @expression(md, z[k=1:p], sum(c[k,j]*x[j] for j=1:n))
julia> @constraint(md, Max, [x[D] for k=1:p])
julia> @constraint(md, [x[D]=b[i], sum(a[D,j]*x[j] for j=1:n) ≤ b[i]]
    
```

Example (1/3)



$$\begin{aligned}\max z_1 &= x_1 + x_2 \\ \min z_2 &= x_1 + 3x_2 \\ \text{s.t.} \quad & 2x_1 + 3x_2 \leq 30 \\ & 3x_1 + 2x_2 \leq 30 \\ & x_1 - x_2 \leq 5.5 \\ & x_1, x_2 \in \mathbb{N}\end{aligned}$$

$$\begin{aligned}\max z_k &= \sum_{j=1}^n c_j^k x_j & k = 1, \dots, p \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i & i = 1, \dots, m \\ & x_j \in \mathbb{N} & j = 1, \dots, n\end{aligned}$$

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julia> @objective(md, Max, [z[k] for k=1:p])
julia> @constraint(md, ct[i=1:m], sum(a[i,j]*x[j] for j=1:n) ≤ b[i])
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Example (1/3)



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Example (1/3)



$$\begin{array}{llll} \max z_1 & = & x_1 & + & x_2 \\ \min z_2 & = & x_1 & + & 3x_2 \\ \text{s.t.} & & 2x_1 & + & 3x_2 \leq 30 \\ & & 3x_1 & + & 2x_2 \leq 30 \\ & & x_1 & - & x_2 \leq 5.5 \\ & & x_1 & , & x_2 \in \mathbb{N} \end{array} \quad \rightarrow \quad \begin{array}{llll} \max z_k & = & \sum_{j=1}^n c_j^k x_j & k = 1, \dots, p \\ \text{s.t.} & & \sum_{j=1}^n a_{ij} x_j \leq b_i & i = 1, \dots, m \\ & & x_j \in \mathbb{N} & j = 1, \dots, n \end{array}$$

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```



Setup and optimize

```
julia> set_optimizer(model, ()->MOA.Optimizer(HiGHS.Optimizer))
julia> set_attribute(model, MOA.Algorithm(), MOA.EpsilonConstraint())
julia> optimize!(model)
```

Get X_E and Y_N

```
julia> for i in 1:result_count(model)
julia>     z_opt = [ objective_value(md; result = i)[k] for k=1:p ]
julia>     x_opt = value.(x; result = i)
julia> end
```



Plot Y_N

```
julia> using Plots
julia> Plots.scatter(
julia> [value(z[1]; result = i) for i in 1:result_count(md)],
julia> [-1 * value(z[2]; result = i) for i in 1:result_count(md)];
julia> xlabel = "objective 1",
julia> ylabel = "objective 2",
julia> title = "objective space",
julia> legend = false,
julia> xlims = (0,25),
julia> ylims = (0,25),
julia> aspect_ratio=:equal,
julia> )
```

↳ https://github.com/xgandibleux/EURO_MCDM2026/blob/main/3.examples/2IPim.ipynb